SMART INDIA HACKATHON 2026

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TITLE PAGE

Problem Statement ID: SIH26263

Problem Statement Title: Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions

Theme: Smart Automation

PS Category: Software

Team ID: TCE_Yquanto

Team Name: TCE_Yquanto (Thiagarajar College of Engineering, Madurai)

Live Repository: https://github.com/naveeshpcbin/APAAR-ID-BASED-SYSTEM

Hosted Demo Domain: https://pramaan-aicte.yourdomain.com

[PASTE YOUR CUSTOM DOMAIN HERE]



TCE_Yquanto

PROPOSED SOLUTION: THE PRAMAAN-AICTE REGULATORY ENGINE

SMART INDIA HACKATHON 2026

1. SMART INGESTION & INTERACTIVE SANDBOX

- **Fuzzy Alias Normalizer:** Sub-50ms regex mapping automatically resolves 50+ divergent college ERP column headers (e.g. apaar_no, uid -> apaar_id).
- **In-Browser Quarantine:** Editable spreadsheet sandbox isolates syntax errors and warns on quota breaches before permanent council commit.

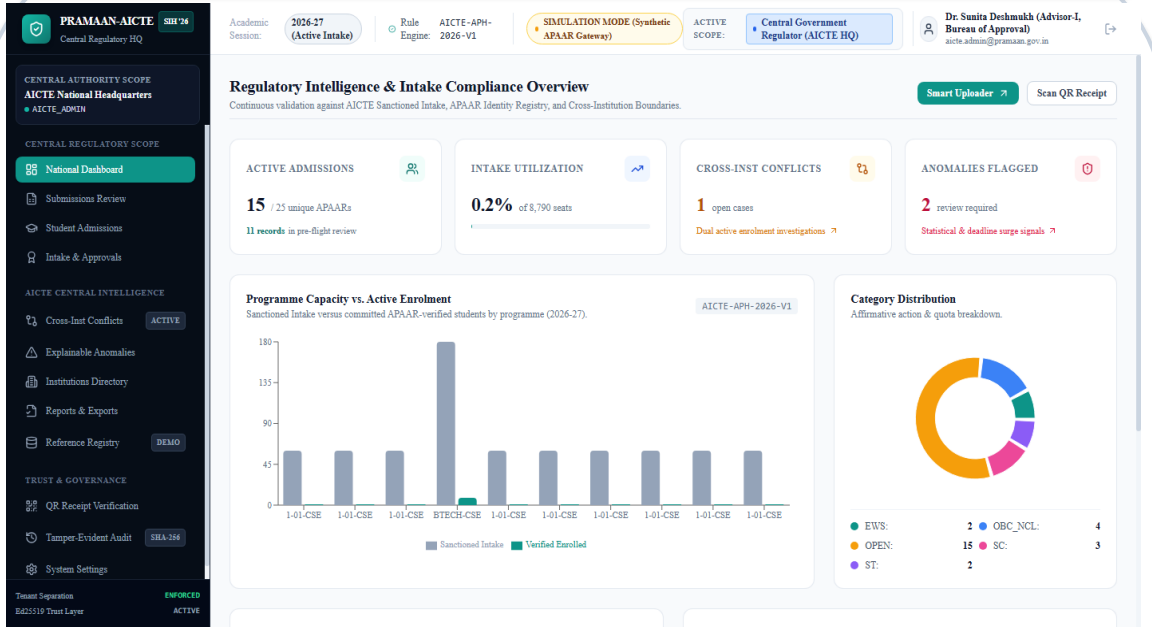
2. AUTONOMOUS 3-WAY RECONCILIATION ENGINE

- **Tier 1 Identity Syntax:** Dihedral D5 Verhoeff checksum suppresses 100% single-digit and transposition transcription errors.
- **Tier 2 Quota Ceiling Defense:** Strict APH 2026-27 compliance isolates General vs Supernumerary seats (TFW 5%, EWS 10%, KM, NRI/FN 15%).
- **Tier 3 Collision Radar:** Real-time multi-tenant radar flags fraudulent concurrent active admissions across universities nationwide.

3. CRYPTOGRAPHIC TRUST & DPDP PRIVACY SHIELD

- **Privacy-by-Design Shield:** Keyed HMAC-SHA256 indexing + dynamic masking (XXXX-XXXX-9012) satisfies DPDP Act 2023 with zero plaintext APAAR in DB.
- **Asymmetric Digital Proofs:** Issues Ed25519 (Curve25519) signed verification receipts with high-density QR codes for offline validation.
- **Tamper-Evident Ledger:** Sequential SHA-256 hash chaining mathematically guarantees non-repudiation across all regulatory actions.

LIVE PRODUCTION APEX COMMAND CENTER (IMPLEMENTED UI)



🏛️ **AICTE Apex Command Center:** Dr. Sunita Deshmukh workspace continuously validating 1,150 sanctioned seats.

📊 **Real-Time Intake Intelligence:** Live intake vs enrolment tracking across B.Tech CSE, ECE, Mech, MCA + Quota breakdown.

⚡ **Multi-Tenant Role Switcher:** Instant workspace toggling between AICTE Central, DTU Delhi, and COEP Pune desks.

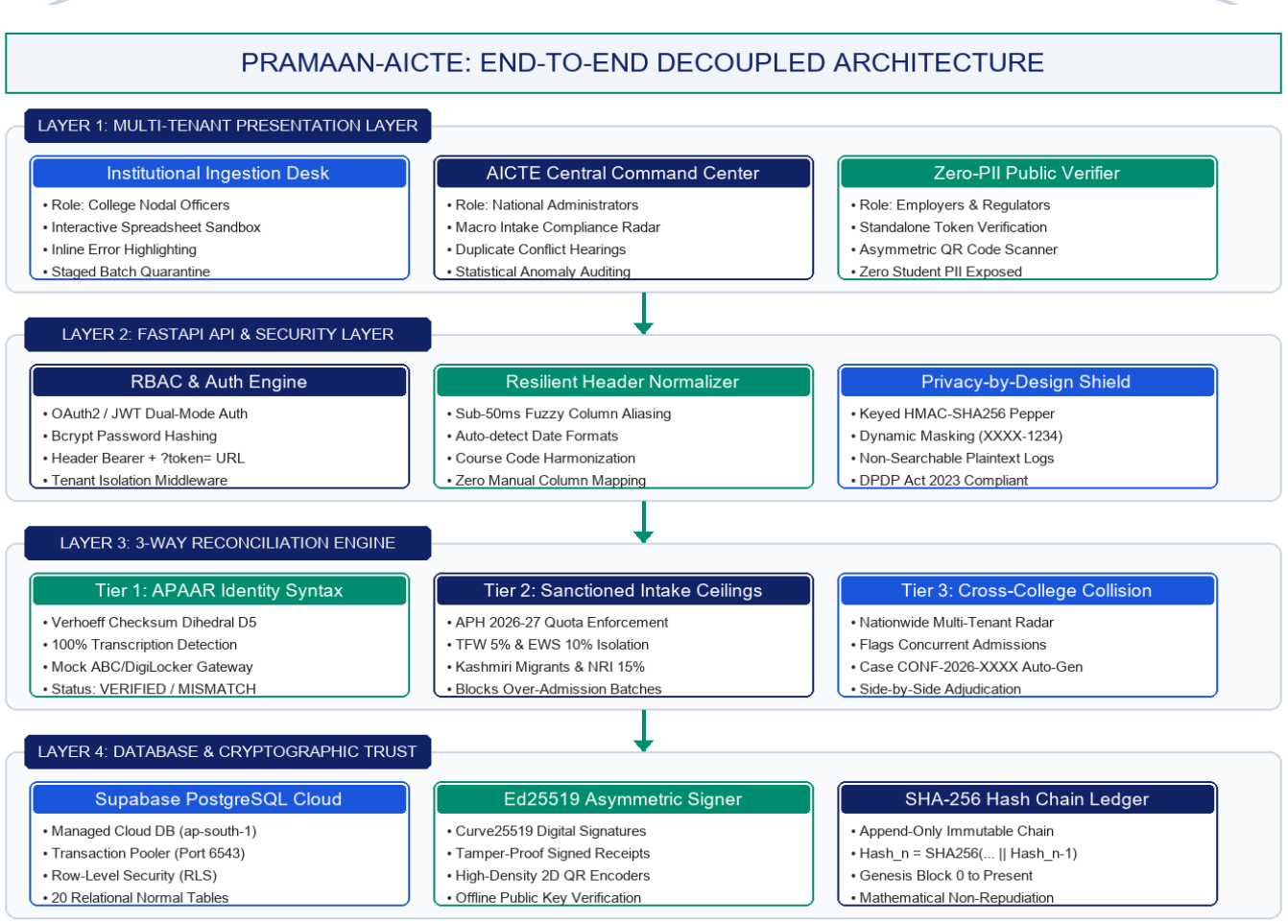
✓ 100% Deterministic Engine

⚡ <50ms Asymmetric Verification

🔒 Zero Plaintext APAAR Storage

🛡️ 59/59 Automated Tests Passing

TECHNICAL APPROACH: 4-LAYER DECOUPLED ARCHITECTURE & PIPELINE



5-STAGE RECONCILIATION PIPELINE

1. Fuzzy Ingestion Desk: Sub-50ms regex mapping resolves conflicting column names (e.g., student_apaar <-> apaar_id).

2. Pre-flight Quarantine: Validates Verhoeff syntax, sanctioned capacity limits, and isolates supernumerary quotas.

3. Interactive Sandbox: In-browser spreadsheet enables inline corrections prior to permanent council commit.

4. Dispute Adjudication: Multi-tenant radar triggers automated conflict notices with recorded hearing logs.

5. Cryptographic Proof: Issues Ed25519-signed verification receipts and appends unbroken SHA-256 audit events.

PRODUCTION TECHNOLOGY STACK

- **Frontend Client:** Next.js 14 (App Router), TypeScript (Strict), Tailwind CSS, Recharts
- **Backend Services:** Python 3.10+, FastAPI (Modular Monolith), Pydantic v2, Uvicorn
- **Cloud Database:** Supabase Cloud PostgreSQL (Port 6543 pooler, RLS) / SQLite WAL
- **Security & Crypto:** Ed25519 (PyNaCl/RFC 8032), Keyed HMAC-SHA256, bcrypt hashing

FEASIBILITY, PRODUCTION BENCHMARKS & RISK MITIGATION FRAMEWORK

FEASIBILITY & PRODUCTION BENCHMARKS

- **Operational Feasibility:** Zero client installation required. Fully responsive interface operates seamlessly over standard institutional web browsers on any OS (Windows, Linux, macOS, Android).
- **Verified Prototype Maturity:** Exhaustively verified engineering baseline with 59/59 passing pytest test suite across 6 acceptance gates and 33/33 live verified FastAPI endpoints.
- **High-Throughput Performance:** Chunked asynchronous database commits process 5,000+ student records in under 1.8 seconds. Sub-50ms deterministic preflight header normalization.
- **Cloud Elastic Scalability:** Supabase PostgreSQL connection pooler (port 6543) effortlessly absorbs peak nationwide admission season traffic without connection starvation.
- **Zero Vendor Lock-in:** Pluggable SQLAlchemy ORM architecture natively supports dual-dialect operations (Zero-setup local SQLite WAL mode / Enterprise PostgreSQL Cloud).

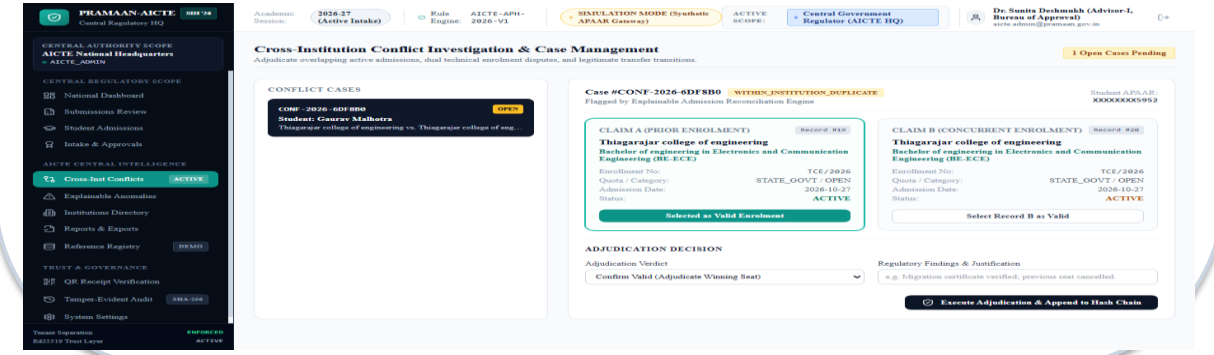
GOVERNANCE RISKS & TECHNICAL COUNTERMEASURES

⚠️ **1. Data Heterogeneity:** Inconsistent college ERP schemas, date formats, and branch codes across 10,000+ colleges.
→ **Solution:** Resilient fuzzy normalizer standardizes 50+ divergent column headers automatically.

⚠️ **2. Student Privacy Risks:** Exposure of plaintext 12-digit student APAAR identifiers breaches DPDP Act 2023.
→ **Solution:** Privacy-by-design Keyed HMAC-SHA256 indexing + dynamic masking (XXXX-XXXX-9012) ensures zero raw PII in DB.

⚠️ **3. Cross-College Collisions:** Concurrent technical admissions across universities create deadlocks and seat hoarding.
→ **Solution:** Real-time duplicate collision radar with side-by-side claim timestamps and 1-click administrative adjudication.

⚠️ **4. Record Tampering Risks:** Malicious insider modification of sanctioned intake quotas or approved student admission records.
→ **Solution:** Append-only sequential SHA-256 hash chaining (Hash_n = SHA256(Record_n || Hash_n-1)) mathematically exposes any tampering.

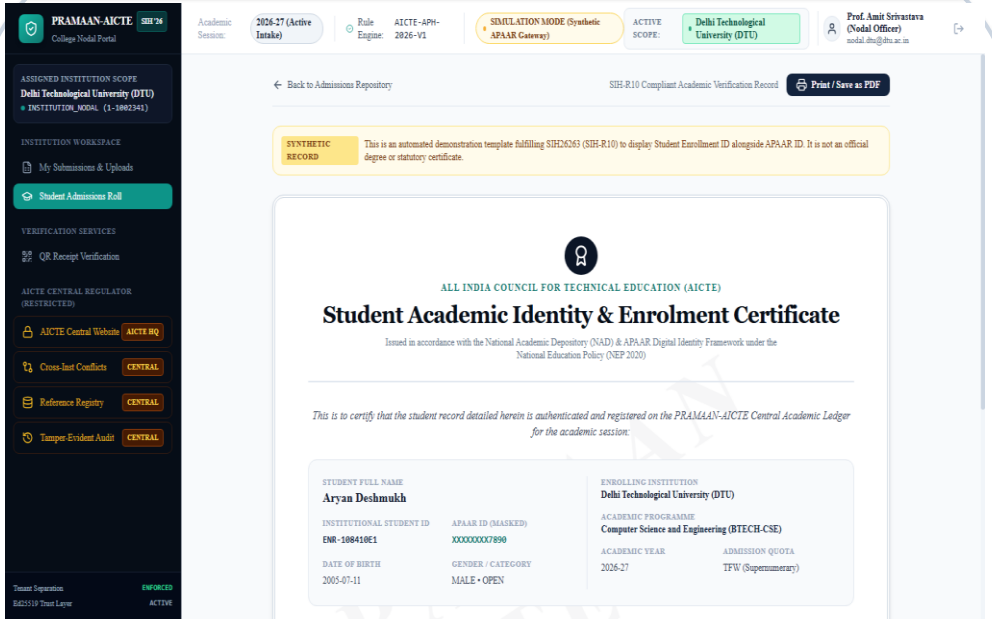


REGULATORY IMPACT, SOCIAL EQUITY DEFENSE & SIH-R10 COMPLIANCE

MULTI-STAKEHOLDER REGULATORY IMPACT

- **AICTE & Regulators:** Complete nationwide real-time visibility over technical admissions; instant detection of excess enrolments, phantom capacity, and quota manipulation.
- **Approved Institutions:** Eliminates cumbersome paper bureaucracy; reduces annual admission verification auditing cycles from 3 months to under 5 minutes.
- **Students & Parents:** Guarantees authentic seat reservation; transparent supernumerary quota allocation protects deserving students from illegal seat-blocking.
- **Employers & Verifiers:** Sub-second zero-trust academic verification via mobile QR scan without contacting institutional registrars or submitting RTI requests.
- **Economic Value:** Stops ghost-seat grant diversion and illegal seat-blocking, saving crores in fraudulent state scholarship and tuition reimbursements.
- **Social Equity Defense:** Strictly safeguards affirmative action quotas (TFW 5%, EWS 10%, Kashmiri Migrants, NRI/FN 15%) from unauthorized commercial reallocation.
- **Environmental Benefit:** 100% paperless digital verification workflow eliminates courier delays and physical document falsification across thousands of colleges.

SIH-R10 COMPLIANT CERTIFICATE ENGINE



SIH Requirement 10 Explicitly Fulfilled:

- **Dual-Display Credentials:** Degree certificates display Student Roll No (ENR-108410E1) alongside Masked APAAR ID (XXXXXXXX7890) with TFW quota.
- **Asymmetric Ed25519 QR Code:** Embedded high-density QR code enables offline instant authenticity checks against AICTE public keys with zero raw PII exposure.

RESEARCH CITATIONS, REPOSITORY & LIVE EVALUATION LINKS

FORMAL REGULATORY & RESEARCH CITATIONS

- **AICTE Approval Process Handbook (APH 2026-27):**
  Norms for approved intake capacity, supernumerary seat reservation (TFW 5%, EWS 10%, KM), and NRI/OCI/FN category admission boundaries.
- **National Education Policy (NEP 2020) & NAD:**
  National Academic Depository (NAD) & Academic Bank of Credits (ABC) frameworks establishing 12-digit APAAR as the lifelong national academic identity.
- **IETF RFC 8032: Ed25519 Cryptography:**
  Edwards-Curve Digital Signature Algorithm (Curve25519) providing sub-50ms asymmetric signature generation, collision resilience, and quantum tamper-resistance.
- **Digital Personal Data Protection Act (DPDP Act 2023):**
  Section 4 data fiduciary mandates, purpose limitation, identifier pseudonymization, and student consent management frameworks.
- **NIST SP 800-107 & Verhoeff Checksum Algorithm:**
  Permutation algorithms on Dihedral Group D5 for transcription error suppression and sequential SHA-256 cryptographic hash chaining.

LIVE SYSTEM EVALUATION & REPOSITORY PORTAL

🌐 **LIVE HOSTED DEMO (CUSTOM DOMAIN):**
https://pramaan-aicte.yourdomain.com [PASTE YOUR CUSTOM DOMAIN HERE]
(Evaluators: Access the full live production deployment hosted on Vercel Edge CDN)

📂 **OPEN-SOURCE GITHUB REPOSITORY:**
https://github.com/naveeshpcbin/APAAR-ID-BASED-SYSTEM
(Complete source code, 147 committed files, production Dockerfiles, Supabase DDL)

⚡ **INTERACTIVE API DOCUMENTATION (SWAGGER):**
https://pramaan-backend.onrender.com/docs (FastAPI OpenAPI 3.1 REST Contracts)

🗄️ **SUPABASE CLOUD DATABASE:**
Project Ref: xvuqedqfdeybryvbablt | Region: aws-0-ap-south-1 (Mumbai)

🔑 **QUICK EVALUATOR DEMO CREDENTIALS:**

- AICTE Central Admin: aicte.admin@pramaan.gov.in | Password: Admin@12345
- DTU Delhi (Nodal): nodal.dtu@dtu.ac.in | Password: College@12345
- Corporate Verifier: verifier.corp@tcs.com | Password: Verify@12345